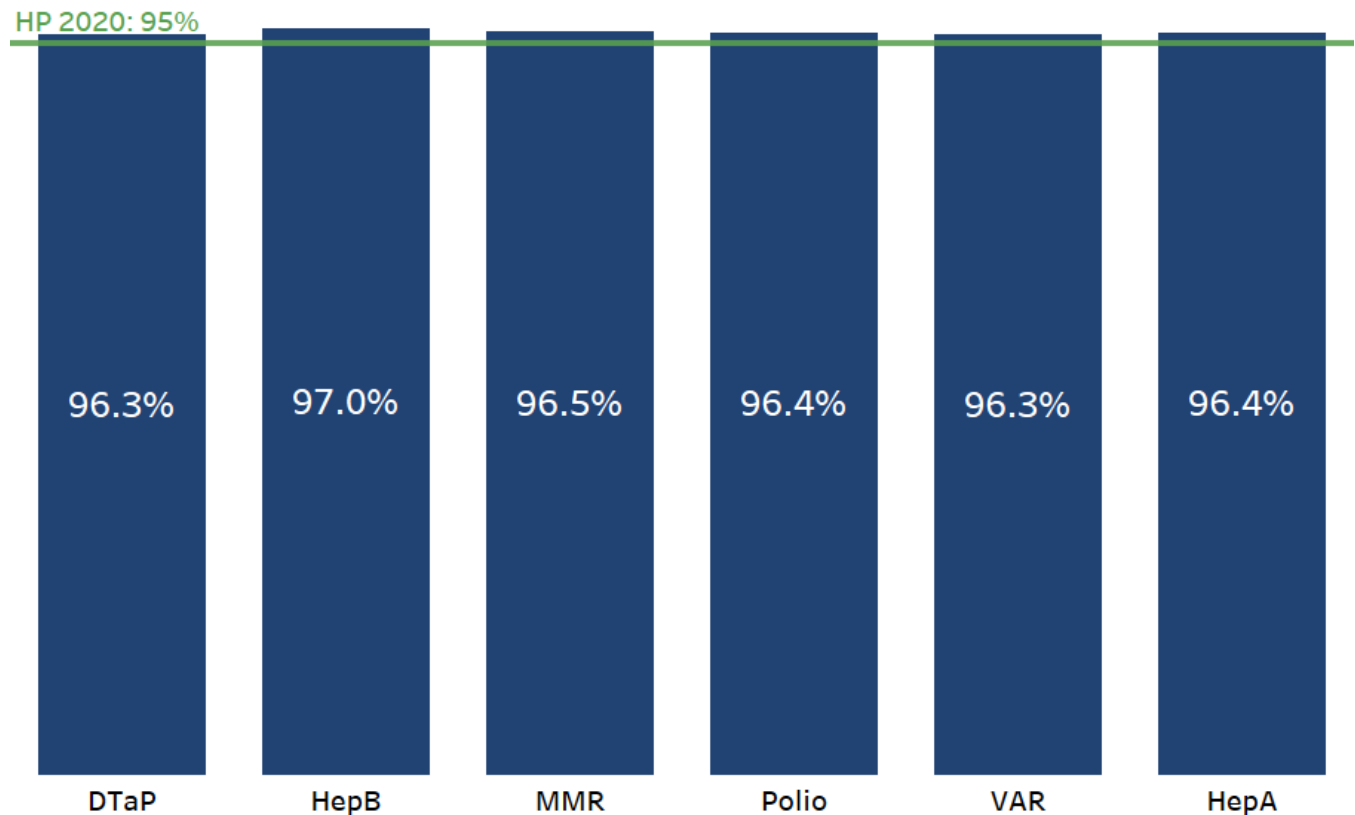


Figure 1. Percentage of public and private school kindergarten students who were complete for each vaccine series, compared to Healthy People 2020 (HP 2020) goal of 95% complete



Public Schools Statewide Summary

Of the 1,049 public schools that were asked to complete the 2020-2021 Kindergarten Assessment 1,010 (96%) submitted results. Statewide, 95.4% of public school students were fully immunized at the time of the survey (66,760 of 69,980 students). Of all public school students, 71.2% (n=49,849) had their immunization record completed and validated in TennIIS. Students who were not fully immunized fell into the following categories, in decreasing order of prevalence:

- Religious Exemption: 1.7% (n=1,177)
- Incomplete Record: 1.2% (n=870)
- Temporary Certificate: 1.0% (n=706)
- Missing Record: 0.5% (n=322)
- Medical Exemption: 0.1% (n=84)
- Transfer <30 days: 0.1% (n=62)

Private Schools Statewide Summary

A total of 176 private schools (out of 295 schools; 61%) with kindergarten enrollment completed the 2020-2021 Kindergarten Assessment. This represents a 10% decline in school participation compared to that of the 2019-2020 Kindergarten Assessment when 71% of private schools participated. We suspect resource burdens caused by the SARS-CoV-2 pandemic contributed to this decline. Statewide, 92.3% (3,550 of 3,848 students) of private school kindergarten students were fully immunized, which represents a 1.5% decrease in the fully-immunized rate from the previous year. Unlike the public school 2020-2021 Kindergarten Assessment, the private school

survey does not ascertain how many private school students had immunization certificates that were validated by TennHIS. Students who were not fully immunized fell into the following categories, in decreasing order of prevalence:

- Religious Exemption: 4.0% (n=155)
- Incomplete Record: 2.3% (n=90)
- Missing Record: 1.0% (n=37)
- Temporary Certificate: 0.7% (n=27)
- Medical Exemption: 0.1% (n=3)
- Transfer <30 days: N/A for private schools based on data collection methods

Trends over Time

Kindergarten student enrollment decreased by nearly 10% in 2020-2021, resulting in 73,828 total students surveyed in the Kindergarten Assessment. Enrollment shifted slightly towards private schools, with approximately 5.2% of the state's total kindergarten enrollment being in private schools compared to 4% last year. The percentage of public and private school students who received all vaccines per Tennessee's state requirements for kindergarten entry (i.e., were "fully immunized") has consistently been close to 95% over the past five years (range: 94.8% to 95.3%). For a community to remain at low-risk for a highly infectious vaccine-preventable disease outbreak such as measles or pertussis, this rate must remain above 95%. Although Tennessee continues to rank just above this threshold statewide, there has been a substantial decline in the number of counties that meet this minimum threshold for vaccine-preventable disease immunity. In fact, the number of counties with public schools that meet the minimum threshold of 95% fully-immunized threshold has decreased from 87 to 72 counties representing a decrease of 5.3%. Overall, the number of exemptions has remained relatively consistent over the past five years. For the past four years, temporary records have increased slowly while incomplete records have declined proportionally. This year, these two categories reversed course with temporary certificates decreasing by 0.9% and incomplete records increasing by 0.5%. Although both record types include children who are missing vaccinations, temporary certificates expire one month after the date that the next required immunization(s) is(are) due which indicates they are at least on track to being fully-immunized. We suspect this uncharacteristic trend to be the result of school shutdowns and administrative burden caused by the pandemic. Regardless of the cause more incomplete records indicates unvaccinated children have fallen behind and need to catch-up on their immunizations. Furthermore, the increasing number of kindergarteners who are not fully-immunized indicates Tennessee schools and communities may be more susceptible to vaccine-preventable disease outbreaks.

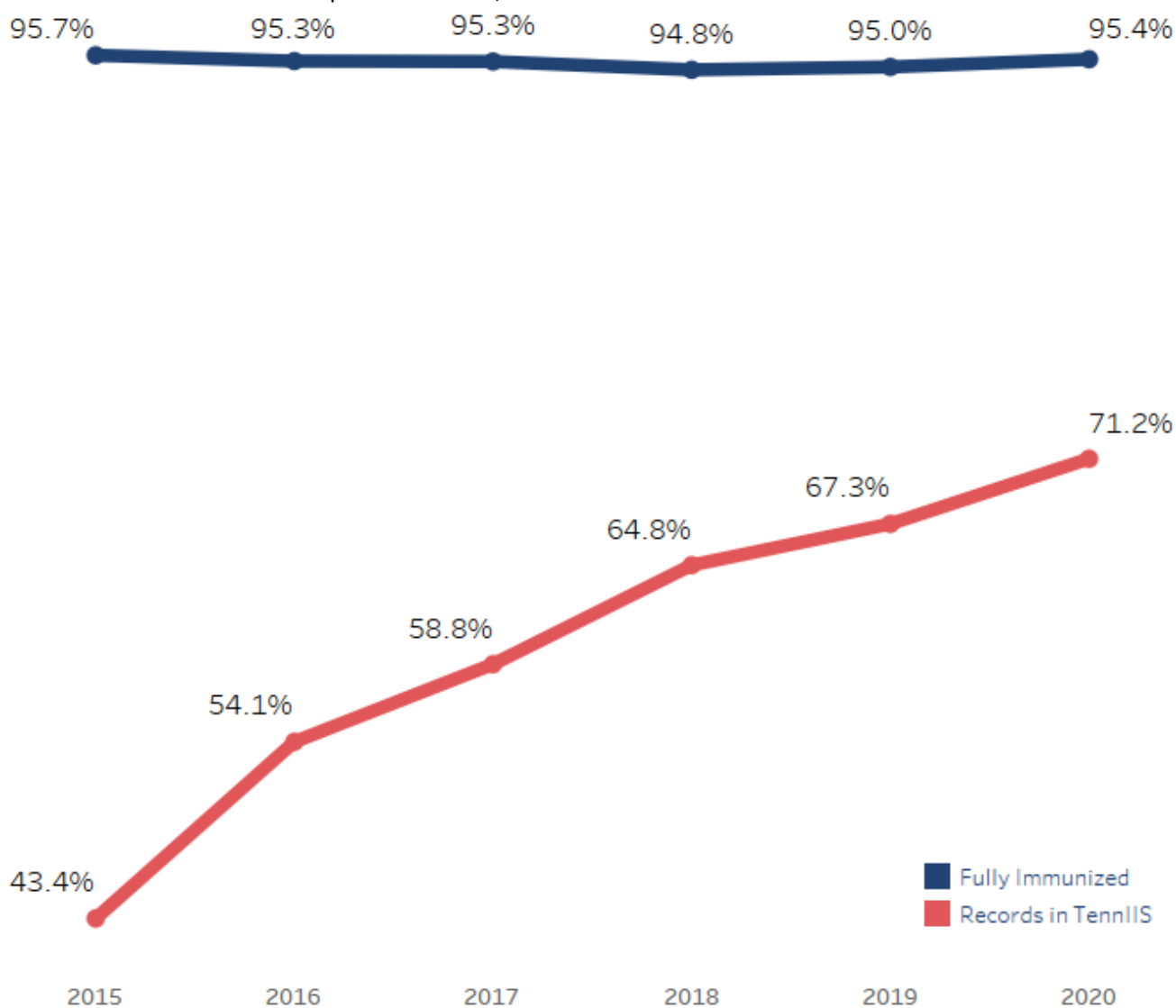
Impact of COVID Pandemic

The CDC noted a decline in overall vaccination rates as soon as one week after the United States declared a national state of emergency due to the SARS-CoV-2 pandemic. The 2020-2021 Kindergarten Immunization Compliance Assessment Report was conducted from November 2020 to February 2021 and affords a unique opportunity to document the impact of the pandemic upon childhood immunization rates in Tennessee. While the overall decline in enrollment was almost certainly the result of pandemic restrictions it is unclear if the same forces contributed to the slight (1.2%) shift in enrollment towards private schools. More granular analysis of the 2020-2021 data showed an 8% increase in counties with public schools that fall below the 95% fully-immunized threshold. This is concerning because only 38% of counties with private school responses exceed 95% of students fully immunized and another 38% failed to achieve even 90% of their students fully immunized. Furthermore, the number of religious exemptions in private schools rose by 1.3% while public school religious exemptions experienced almost no change. These communities are at increased risk of vaccine-preventable disease outbreaks. Data for this report does not reflect the full impact of the pandemic upon kindergarten

immunization rates. Immediate action to increase compliance with recommended childhood immunizations is necessary to decrease the risk of future vaccine-preventable disease outbreaks in our schools in communities.

The percentage of public school students with immunization records documented in TennIIS rose from 43.4% in the 2015-2016 school year to 71.2% in the 2020-2021 school year: a 28% increase in just five years. This significant increase is attributed to the efforts of the VPDIP program to increase TennIIS participation among immunizing providers and school nurses statewide.

Figure 2. Percentage of public-school students fully immunized, and percentage of public school students whose immunization records were complete in TennIIS, over time



The percentage of all TN kindergarten students who were not fully immunized, by category, over time, are shown in **Figure 3**. Over the past five years, the percentage of students with an incomplete immunization record decreased until 2020. (see Figure 3, gold line). This change may be due to disruptions caused by the SARS-CoV-2 pandemic, which created barriers to obtaining immunizations as well as administrative challenges in the schools responsible for enforcing the state's immunization requirements. Until this year, the percentage of children with a temporary immunization certificate had been increasing (see Figure 3, blue line). This change could be due to

High immunization coverage rates are critical to avoiding vaccine-preventable disease (VPD) outbreaks (e.g., measles, pertussis). A VPD outbreak may occur when community members are inadequately immunized against a disease. To remain at low risk for a VPD outbreak the immunization rate should remain at or above 95%. In the maps above and tables below, an immunization rate of at least 95% meets this immunization goal, an immunization rate of 90% to 94.9% needs improvement, and an immunization rate below 90% places the community at increased risk for vaccine-preventable disease outbreaks.

For detailed Kindergarten Assessment results at the county, school districts, and school levels, please see the *Public School Districts (SD) by County Summary* section below.

Table 1. Percentage of **public and private school** students fully immunized, total # students, and the number of schools with students fully immunized within each color category, by county, in descending order by “% students fully immunized” (Color Key: <90%; 90-94.9%; 95-100%)

County	% students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County	# Schools with 90-94% Students in County	# Schools with 95-100% Students in County
LAUDERDALE	100.0%	237	0	0	2
SMITH	99.5%	213	0	0	5
HARDIN	99.3%	276	0	0	6
CROCKETT	99.1%	214	0	0	5
SCOTT	99.0%	296	0	0	6
MARSHALL	98.9%	278	0	0	2
COCKE	98.4%	387	0	2	8
HARDEMAN	98.4%	257	0	1	5
SEQUATCHIE	98.4%	124	0	0	1
DYER	98.3%	460	0	0	6
GREENE	98.2%	609	0	0	12
GRAINGER	98.1%	209	0	0	4
MARION	98.1%	316	0	0	5
BENTON	98.0%	153	0	1	2
TIPTON	97.9%	761	0	0	8
TROUSDALE	97.9%	97	0	0	1
CANNON	97.8%	134	0	2	4
UNICOI	97.7%	132	0	1	3
MACON	97.4%	386	0	1	2
CLAIBORNE	97.3%	329	0	3	5

County	% students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County	# Schools with 90-94% Students in County	# Schools with 95-100% Students in County
DICKSON	97.3%	562	0	2	8
FRANKLIN	97.3%	400	0	3	7
HENRY	97.3%	296	1	1	3
BLEDSON	97.2%	106	0	1	2
HENDERSON	97.2%	326	0	1	7
WARREN	97.2%	425	0	1	8
CUMBERLAND	97.1%	488	0	1	8
LAKE	97.0%	66	0	0	1
HAWKINS	96.9%	482	0	4	8
LINCOLN	96.9%	327	0	2	4
MCNAIRY	96.9%	290	0	0	5
SEVIER	96.9%	1,048	1	4	11
WAYNE	96.9%	159	0	1	2
CARROLL	96.8%	316	0	1	4
ANDERSON	96.7%	828	1	4	12
CHEATHAM	96.7%	459	1	0	6
COFFEE	96.5%	636	0	3	9
DECATUR	96.5%	114	0	0	2
LOUDON	96.5%	372	1	0	6
WILSON	96.5%	1,734	2	3	18
LEWIS	96.4%	137	0	0	1
MADISON	96.4%	995	0	4	12
WASHINGTON	96.4%	1,197	2	1	21
WEAKLEY	96.4%	278	0	2	3
MCMINN	96.2%	526	1	2	7
OVERTON	96.1%	205	0	2	3
RUTHERFORD	96.1%	4,257	1	17	26
SHELBY	96.1%	11,465	12	37	133
SULLIVAN	96.1%	1,337	2	6	18

County	% students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County	# Schools with 90-94% Students in County	# Schools with 95-100% Students in County
SUMNER	96.0%	2,123	3	5	24
MEIGS	95.9%	123	0	0	2
JEFFERSON	95.8%	481	0	3	5
OBION	95.8%	385	0	2	4
HICKMAN	95.7%	230	0	1	1
DEKALB	95.6%	228	0	0	2
CAMPBELL	95.5%	374	1	3	5
HUMPHREYS	95.5%	220	0	0	3
GIBSON	95.4%	669	1	2	7
HAMBLEEN	95.1%	771	2	3	8
WILLIAMSON	95.1%	3,035	5	11	28
BEDFORD	95.0%	624	0	4	5
FAYETTE	95.0%	278	0	3	3
ROBERTSON	95.0%	906	1	4	8
RHEA	94.8%	386	0	1	4
CARTER	94.7%	550	2	4	6
GRUNDY	94.7%	131	1	2	3
KNOX	94.7%	4,716	10	22	44
MONTGOMERY	94.7%	2,820	2	13	13
BLOUNT	94.3%	1,362	4	6	10
HOUSTON	94.3%	87	0	1	1
DAVIDSON	94.0%	6,907	21	25	61
LAWRENCE	93.9%	477	1	4	5
JOHNSON	93.8%	129	0	3	1
CHESTER	93.2%	207	0	1	2
MOORE	93.2%	59	0	1	0
MAURY	93.0%	1,104	4	6	8
MONROE	93.0%	471	0	4	2
MORGAN	93.0%	200	1	0	4

County	% students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County	# Schools with 90-94% Students in County	# Schools with 95-100% Students in County
STEWART	93.0%	128	0	2	0
PUTNAM	92.6%	797	3	6	3
ROANE	92.6%	433	3	2	2
CLAY	92.5%	80	0	1	1
POLK	92.3%	130	1	1	2
GILES	92.1%	290	0	4	0
PICKETT	92.0%	50	0	1	0
HANCOCK	91.7%	60	0	1	0
HAMILTON	91.6%	3,689	21	18	22
BRADLEY	91.3%	1,114	8	9	4
FENTRESS	90.7%	172	1	3	0
PERRY	90.0%	70	1	0	2
WHITE	89.4%	302	3	1	2
JACKSON	88.8%	107	1	1	0
UNION	85.3%	401	4	2	0
VAN BUREN	84.7%	59	1	0	0
HAYWOOD	84.0%	194	1	0	0

Public and Private School Summaries by County and School District

The following series of tables and graphs capture information at two different levels: 1) at the county-level, featuring information for both public and private schools within the county; and 2) at the school district level, featuring information for all schools within the school district. Data are presented by county in alphabetical order, with school districts nested under each respective county in alphabetical order, and schools nested under each respective school district in alphabetical order.